

Department of Electrical and Computer Engineering
University of North Carolina at Charlotte
ECGR 4105/5105: Introduction to Machine Learning

Summer 2026 Midterm Project Description Prof. A. Arafa & TA: A. Ali

1 Groups

Students will be divided into groups of **three (3)**. Submit the list of members in your group via email to TA: aali28@charlotte.edu (and CC: aarafa@charlotte.edu).
Due: Sunday 7/12.

2 Description

Using the Breast Cancer Dataset, you are required to implement and evaluate **at least five (5)** of the following classifiers:

1. Logistic Regression
2. Naive Bayes
3. Perceptron
4. Support Vector Machine, SVM
5. Decision Tree Classifier (*new*)
6. K-Nearest Neighbors, KNN (*new*)

Requirements.

Your program should:

1. Load the training and testing datasets.
2. Preprocess the data as needed.
3. Normalize or standardize numeric features when required.
4. Train at least five different classifiers.
5. Report and compare the performance of the classifiers.

Evaluation Metrics.

For each classifier, report the following:

- Confusion matrix
- Accuracy

- Precision
- Recall
- F1-score

Deliverables.

Submit the following:

1. Jupyter notebook (.ipynb): complete, runnable, and organized with code, figures, tables, and concise Markdown explanations.
2. A short report containing:
 - The classifiers used
 - The preprocessing steps applied
 - The evaluation results
 - A comparison table of classifier performance
 - A brief discussion of the best-performing classifier
3. The confusion matrix for each classifier on the validation data.

Notes.

- You are allowed to use built-in machine learning functions and libraries.
- You may use libraries such as pandas, numpy, scikit-learn, and matplotlib.
- You should not train on the validation data.
- You should apply the same preprocessing steps to both the training and validation datasets.

Discussion.

Prepare detailed answers for the following questions:

1. Which model performed best overall? Define what you mean by “best” and justify your choice using more than one metric.
2. Which model achieved the highest recall for malignant cases? Why is this metric especially important for this task?
3. How did scaling affect logistic regression, SVM, and KNN/Decision Tree? Explain the observed changes using the way each algorithm works.
4. For the Decision Tree, how did maxDepth affect training and validation/cross-validation performance? What methods can be used to control overfitting?
5. How does a Decision Tree select a feature and split point at each node? Explain the role of an impurity criterion such as Gini impurity or entropy.

6. For KNN, how did you select k ? Explain why a very small k can overfit and why a very large k can underfit.
7. For each of the five classifiers, explain the main assumptions or inductive biases, the types of decision boundaries it can represent, important sensitivities (for example, scaling, noise, outliers, correlated features, dimensionality, or hyperparameters), and situations in which performance may degrade.

3 Grading

This midterm project constitutes 35% of the overall course grade. This will further be divided as follows:

- Deliverables: 20% (same for all group members); **Due: Sunday 7/19**
- Discussion (Monday 7/20 class): 15%; based on the ability to answer questions (may be different for each group member).